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Ancient platform carbonates with well-developed Lofer cyclicity: new candidates for magnetostratigraphy and geodynamically oriented palaeomagnetism

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SUMMARY

Young platform carbonates seem to preserve continuous records of the magnetic field. In contrast, ancient ones often fail to yield a meaningful palaeomagnetic signal. The frequent failure of the palaeomagnetic method in the platform carbonates of the Alpine–Mediterranean zone resulted in an investigation as to what extent the original lithology is responsible for this situation.

The material selected for the study represents Late Triassic platform facies with well-developed Lofer cycles (metre-scale peritidal–subtidal cycles) of the Transdanubian Central Range in Hungary, with evidence for minimal late diagenetic, tectonic or secondary alteration processes.

By selecting an area where the expected declinations for the Late Triassic are rotated in a counterclockwise manner by a significant amount, relative to the present north, we hope to recognize ancient natural remanence on the basis of both consistency and departure from the Earth's present field direction.

For the study, different members of Lofer cycles were drilled at two sections (13 km apart). 71 and 38 samples, respectively, were taken from the two sections and fully oriented in the field.

Using standard palaeomagnetic processing techniques, ancient natural remanence was isolated in all 'A', 'B' and 'C' members of the Lofer cycle, although different members were characterized by different magnetic parameters ('C' members were diamagnetic with very low NRM intensity; 'A' members had positive susceptibility and relatively strong NRM, and 'B' members intermediate properties).

The mean palaeomagnetic directions isolated for the two sections, based on 58 and 30 samples, respectively, are $D=306°$, $I=47°$, $k=43$, $\alpha_{95}=3°$ and $D=300°$, $I=52°$, $k=26$, $\alpha_{95}=5°$ before and $D_c=313°$, $I_c=52$ and $D_c=307°$, $I_c=52°$ after tectonic correction. Both normal and reversed polarities were observed, and there were several polarity reversals, even in short sections. These results suggest that platform carbonates with Lofer cycles are likely to preserve primary remanence, thus they are suitable for magnetostratigraphy studies, if free from karstic cavities and fissures.

The presence of 'A' members is very favourable for the preservation of the original magnetization (since general water circulation is inhibited by these members), but these may represent condensed intervals, thus the sedimentary record in these members is not necessarily complete.

Despite the above limitations and also the possibility of gaps at sequence boundaries, platform carbonates are good candidates for future definition of magnetostratigraphic scales, for example in the Late Triassic; they also promise palaeomagnetic results of considerable tectonic importance, since platform carbonates with Lofer cycles are widespread in the Alpine–Mediterranean zone.

Key words: carbonates, Lofer cycles, palaeomagnetism, Triassic.

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1 INTRODUCTION

It is well known that carbonates have a weak natural remanent magnetization (NRM). Mesozoic carbonates, particularly shallow-water facies, in the Mediterranean region are no exception: they often fail to yield any meaningful palaeomagnetic signal. Normally, a large number of samples must be studied, with the prospect of obtaining positive results from 40 to 50 per cent of these (e.g. Márton & Veljovic 1983, 1987; Márton & Pugliese 1989; Márton, Milicevic & Veljovic 1990; Márton & D'Andrea 1992).

In contrast, young platform and reefal carbonates seem to have preserved continuous records of the geomagnetic field. Thus, polarity zones can be determined in these rocks and magnetostratigraphy can be applied to a range of problems, such as the timing of atoll growth (Aissaoui, McNeill & Kirschvink 1990).

Palaeomagnetic studies of young platform carbonates (McNeil *et al.* 1988; Aissaoui *et al.* 1990; Aissaoui & Kirschvink 1991) have demonstrated that their primary magnetic mineral is fine-grained magnetite of biogenic or early authigenic origin, with unblocking temperatures well below the Curie point.

With a new strategy of deliberately seeking NRM of moderate stability in platform carbonates, the determination of a Late Cretaceous palaeomagnetic pole for Apulia (Márton & Nardi 1994) was made possible. Nevertheless, reliable results were confined to certain horizons.

Several factors may contribute to the success or failure of such studies. The most important may be the initial sedimentological properties, the effects of early and late diagenesis (including dolomitization), karstification and tectonism.

The influence of each of the above factors is estimable under favourable geological conditions, such as exist in the Transdanubian Central Range of Hungary, where sections of Late Triassic age display a wide variety of lithological compositions and are affected to different degrees by diagenetic, tectonic and secondary alteration processes.

These sections are geologically well studied and have the further advantage that the expected declinations in the Transdanubian Central Range for the Triassic depart significantly from those of the Earth's present magnetic field (Márton & Márton 1983). Thus, for fully oriented samples, both the consistency and ancient age of the remanence may be demonstrated.

This paper presents results obtained from two sections showing a high variability of depositional lithology (including well-developed Lofer cycles) and with minimal evidence for post to early diagenetic alteration and tectonic disturbance.

2 CYCLICITY

Cyclicity in carbonate-platform sedimentation was first recognized in the Late Triassic Dachstein Limestones of the Northern Calcareous Alps (Sander 1936). The exact definition, description and modern genetic interpretation of the cycles are a result of the work of Fischer (Fischer 1964), who introduced the term 'Lofer cycles'.

According to Fischer, a complete Lofer cycle shows a deepening-upward trend. It consists of a supratidal 'A', an intertidal 'B' and a subtidal 'C' member. Member 'A' is represented by argillaceous intraclastic mudstone (red to green) of soil origin (*in situ* or reworked); member 'B' appears as laminated carbonate containing abundant fenestral pores (algal-mat facies as a rule); and member 'C' is massive calcarenitic wackestone or packstone with rich marine biota. The cycle is bounded by weathered unconformable surfaces.

Later, shallowing-upward cycles were also discovered in the Late Triassic Dolomia Principale of the Southern Alps (Bosellini 1965, 1967; Bosellini & Hardie 1985).

Symmetric Lofer cycles (A–B–C–B') were first described from the Norian–Rhaetian Dachstein Limestone of the Bakony Mountains (Haas 1982, 1991). Haas suggested that this cycle type be regarded as standard, since all the observed stacking patterns can be derived from a symmetric cycle (Haas 1982). Schwarzacher & Haas (1986) referred to the Lofer cycles defined above as basic cycles and Strasser (1991), using the terminology of sequence stratigraphy, described them as small-scale depositional sequences. The most probable duration of a basic Lofer cycle is about 20 ka (Schwarzacher & Haas 1986), which corresponds to the shortest Milankovitch period.

Basic Lofer cycles are bounded by disconformity surfaces, which were produced by erosion, dissolution and soil formation. The duration of the subaerial period of a cycle of 20 ka is estimated as a few thousand years. However, the true gaps can be larger, since erosion and dissolution may eliminate part of the formerly deposited sediments during subaerial exposure. Erosion surfaces related to shore-face migration are common within the basic Lofer cycles, but they are not considered to represent significant hiatuses (Riding & Wright 1981).

The basic cycles are superimposed on higher-order cycles, thus resulting in a 'megacyclic' pattern. Megacycles are bundles of 5–7 basic cycles bounded by an anomalously thick development of the 'A' and 'B' facies. These horizons are called master bedding planes (Schwarzacher 1954).

Vail *et al.* (1977, 1991) and Vail (1992) introduced the terminology for subdividing sedimentary successions on the basis of the duration of the rhythmic genetic processes. The longest, first-order cycles (duration >50 Ma) are regarded as tectono-eustatic in origin, i.e. caused by changes in the volume of the oceans due to plate-tectonic processes. The majority of the second-order (3–50 Ma) cycles are believed to be tectonically induced transgressive–regressive facies cycles. The third-order cycles (0.5–3 Ma) are bounded by major eustatic sea-level falls (the amplitude of sea-level change being a few tens of metres).

The fourth- (0.08–0.5 Ma), fifth- (0.03–0.08 Ma) and sixth-order (0.01–0.03 Ma) cycles are within the frequency band of the orbitally forced Milankovitch climatic cycles. The amplitudes of the sea-level oscillations in this case are tens of metres in 'ice-house' periods and up to several metres in 'greenhouse' periods.

As Goldhammer, Dunn & Hardie (1990) demonstrated, certain characteristics (e.g. thickness and facies) of a short-term cycle are controlled by longer-term (higher-order) sea-level variations due to changes in the accommodation depth. For instance, in the lowstand phase of the 3rd-order cycles, the 4th- to 6th-order cycles are probably condensed and dominated by subaerial facies and the subtidal facies are either very thin or totally absent. In contrast, during periods of sea-level rise (increasing accommodation space) the 4th- to 6th-order cycles are generally thick, complete and subtidal facies are prevalent.

3 GEOLOGY AND SAMPLING

According to the majority of authors, the Transdanubian Central Range was situated somewhere between the Southern

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Alps and the Northern Calcareous Alps (Fig. 1) in the early periods of Alpine evolution.

Carbonate-platform sedimentation began in the Transdanubian Central Range area during the Anisian. In some places it was interrupted by a Mid-Triassic mobile episode when deeper intraplatform basins were formed. Equilibration of the dissected topography was complete by the Late Carnian as a result of terrigenous infilling of the basins.

From the latest Carnian to the end of the Rhaetian, a 2–2.5 km thick platform-carbonate sequence accumulated on a very wide shelf. At the beginning of this evolution (late Carnian to middle Norian), diagenetic dolomitization took place under arid conditions (the main dolomite stage) due to the moderate subsidence rate. Later, accelerated subsidence and other factors (e.g. a more humid climate) led to the formation of a rimmed platform where the practically undolomitized Dachstein Limestone, of 800–1000 m thickness, was formed (late Norian to Rhaetian). Both the macrofauna (*Megalodon*) and the microfauna (*Foraminifera*) justify such subdivision.

Within the Dachstein Formation, two main facies types can be distinguished. Oncoidal development characterizes the outer shelf zone, whereas the inner platform exhibits Lofer cycles. The latter are exposed in outcrops and quarries in the Gerecse and Bakony Mountains (Fig. 2).

For palaeomagnetic sampling, two large quarries at Lábatlan and Agostyán were selected in the Gerecse Mountains (Fig. 2) both in sediments of Rhaetian age. Apart from the causes mentioned in the Introduction, these sections were good candidates for palaeomagnetic work because of their clearly observable higher-order cyclicity. Palaeomagnetic sampling sites and their members and lithologies are shown in Figs 7 and 8.

Approximately 110 oriented samples were drilled from different members of the Lofer cycles. The cores were oriented in the field using a magnetic compass. The respective

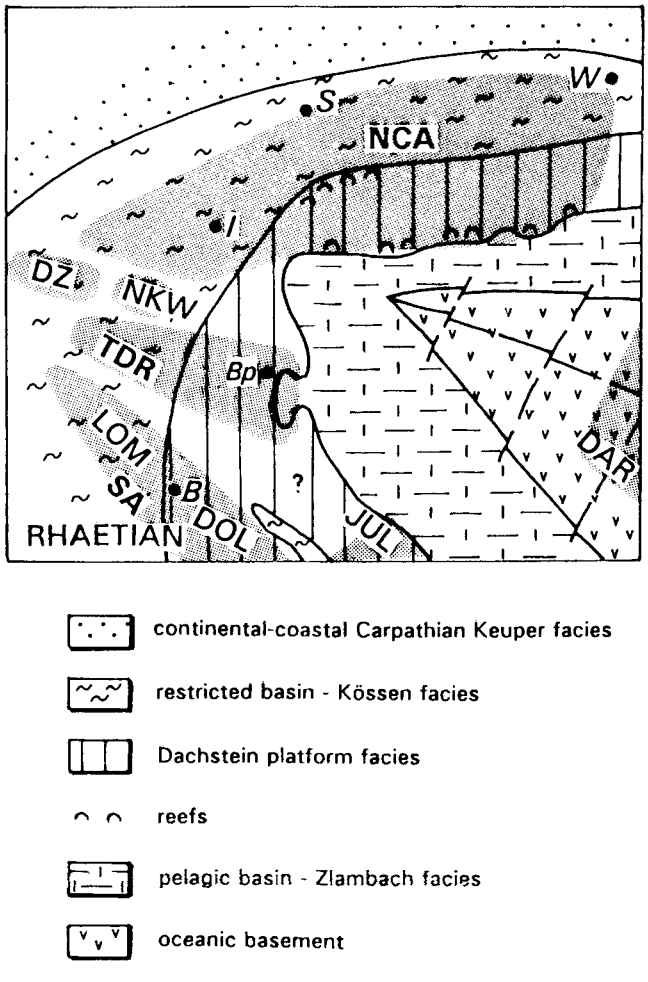


Figure 1. Palaeogeographic setting of the study area in the Rhaetian. NCA: Northern Calcareous Alps; DZ: Drauzug; NKW: Northern Karawanks; TDR: Transdanubian Range; DAR: Darno Unit (N. Hungary); SA: Southern Alps; LOM: Lombardy; DOL: Dolomites; JUL: Julian Alps; W: Vienna; S: Salzburg; I: Innsbruck; Bp: Budapest; B: Bolzano.

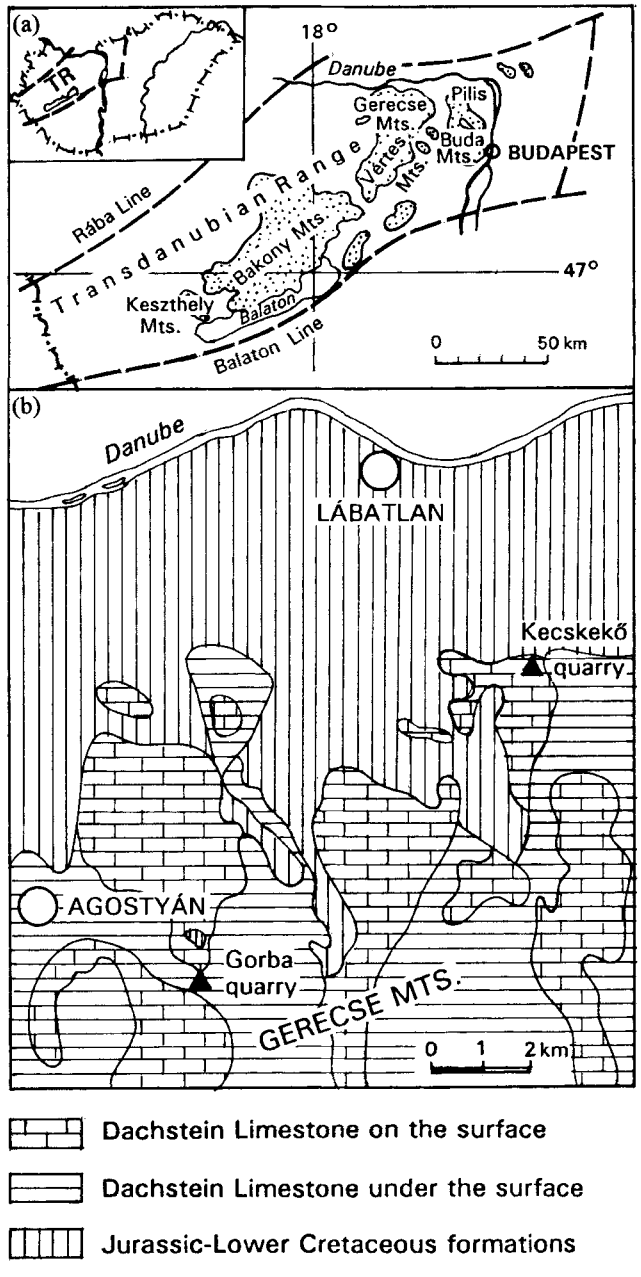


Figure 2. (a) Location of the Transdanubian Range (TR) in Hungary and the position of the Gerecse Mountains. (b) Location of the studied sequences.

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thicknesses of the stratigraphical sequences sampled are 5.5 and 2.4 m. Thus, relative to the estimated average sedimentation rate (12.5 cm ka^{-1}) the sampling was dense.

4 LABORATORY MEASUREMENTS

The oriented samples were cut into standard-sized specimens and then subjected to measurements in the Palaeomagnetic Laboratory of the Eötvös Loránd Geophysical Institute of Hungary. First the NRM and the susceptibility of the specimens were measured in their natural state on a two-component cryogenic magnetometer, on JR-4 spinner magnetometers and on a KLY-2 susceptibility bridge. Several pilot samples (about one-third of the total of 110) were then subjected to AF demagnetization in moderate AF fields (maximum 10 mT). The demagnetization was continued on the same specimens using the thermal method, in increments of 25–50 °C, starting with 150 °C until the signal was lost (the noise level for measurement of NRM is 6×10^{-6} A m^{-1}) or up to 575 °C. In order to monitor changes in magnetic mineralogy, susceptibility was remeasured after each heating step.

Occasionally, AF demagnetization in high fields (120–235 mT) was applied after 575 °C. This was carried out to demonstrate convincingly the contribution of haematite to the NRM, since the thermal demagnetization above 575 °C was impractical due to rapidly increasing susceptibility. All other 'non-pilot' samples were also completely demagnetized in several steps, but fewer steps than the pilots.

In order to identify the magnetic minerals present in a sample, IRM acquisition experiments using a pulse magnetizer and an electromagnet (peak fields 1.0 and 2.4 T, respectively) were carried out. Sometimes this was followed by thermal demagnetization of the three-component IRM (Lowrie 1990), the hard component being acquired in a field of 2.4 T, the medium component in a field of 0.36 T and the soft component in a field of 0.16 T.

5 RESULTS AND DISCUSSION

As expected, both the NRM intensity and the susceptibility values were very weak in the sampled sections. Samples from undisturbed 'C' members, which are typical of the Lábatlan Kecskekő quarry section (Fig. 2) invariably had diamagnetic susceptibilities. In 'A' and 'B' members the susceptibility was

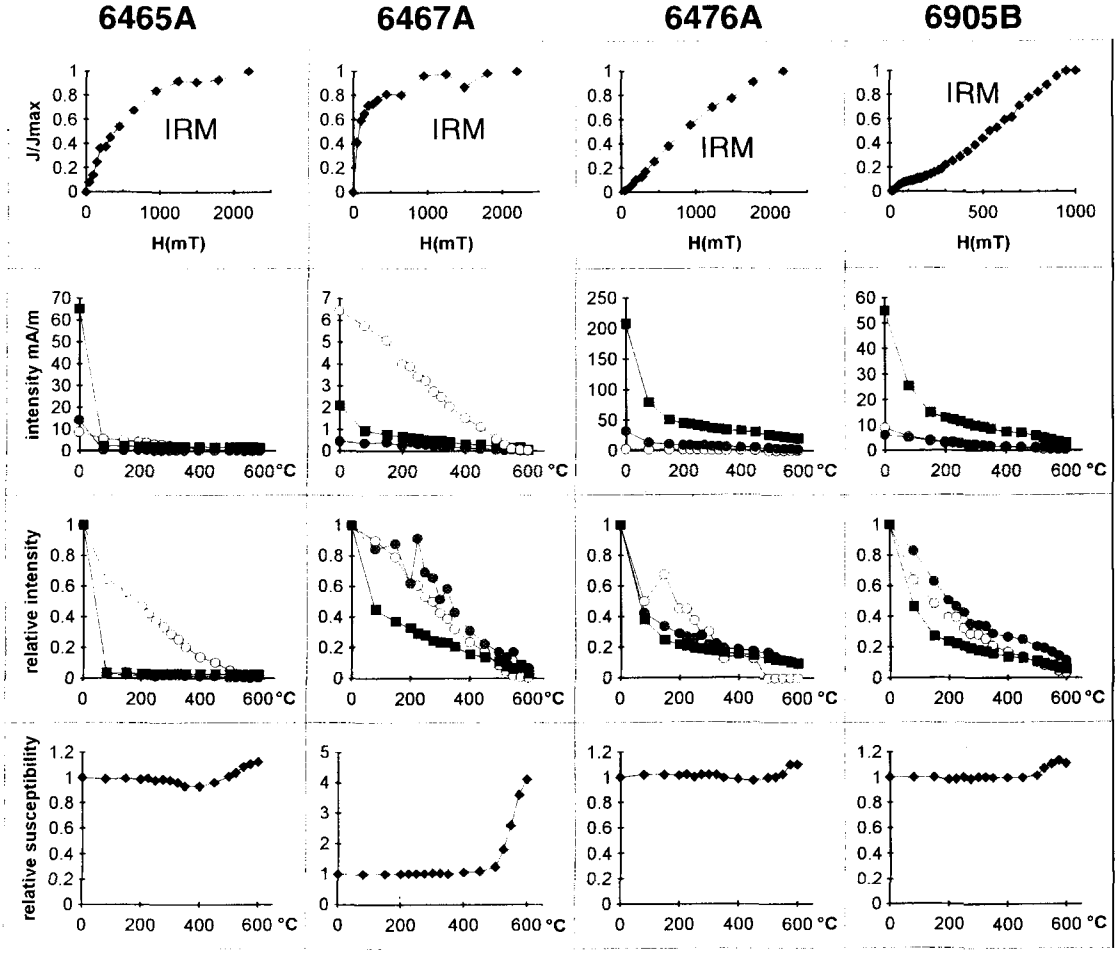


Figure 3. Magnetic identification of the potential carriers of the NRM. From top to bottom: IRM acquisition curves; intensity of the three-component IRM (soft: open circles; medium: solid circles; hard: squares) on thermal demagnetization; same normalized to the initial values; normalized susceptibility measured after each heating step. 6465A, 6467A and 6905B are normal-polarity samples and 6476A are reversed-polarity samples.

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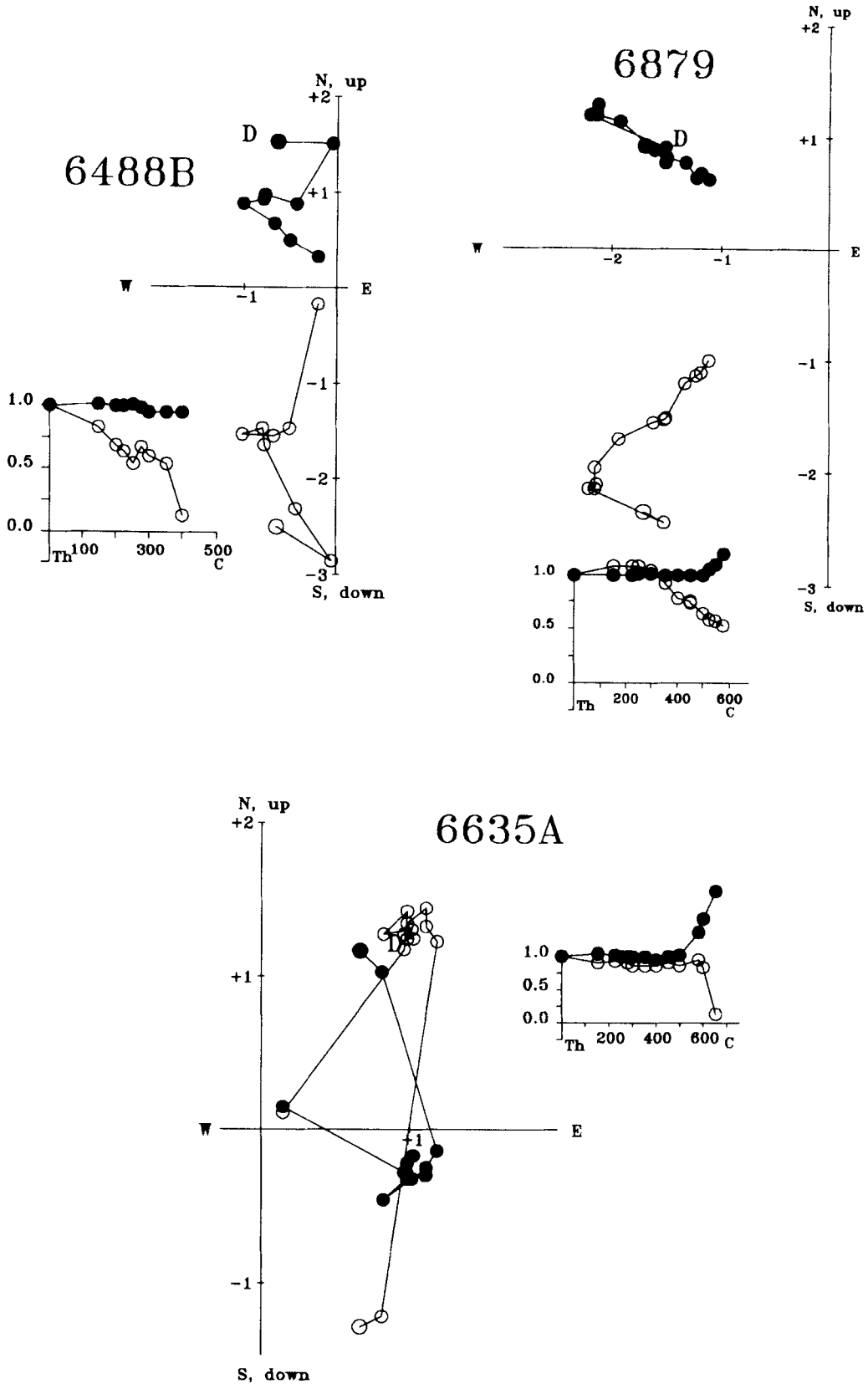


Figure 4. Typical behaviour of the NRM and susceptibility on demagnetization. Demagnetization steps are 10 mT (AF) followed by thermal demagnetization [steps can be read from the relative intensity (open circles) and susceptibility (solid circles) versus temperature curves]. Decay of the NRM vector is shown in orthogonal plots (solid circle: horizontal plane; open circle: vertical plane; large circles represent NRMs). One unit on the orthogonal plots: 8×10^{-6} A m^{-1} (6488B Lábatlan 'C' member), 1.6×10^{-4} A m^{-1} (6879 Lábatlan 'B' member), 4×10^{-5} A m^{-1} (6635A Agostyán 'C' member), 4×10^{-5} A m^{-1} (6467A and B Lábatlan 'A' member; 6467A only AF demagnetization; 6467B thermal demagnetization), 1.6×10^{-4} A m^{-1} (6880 Lábatlan 'B' member).

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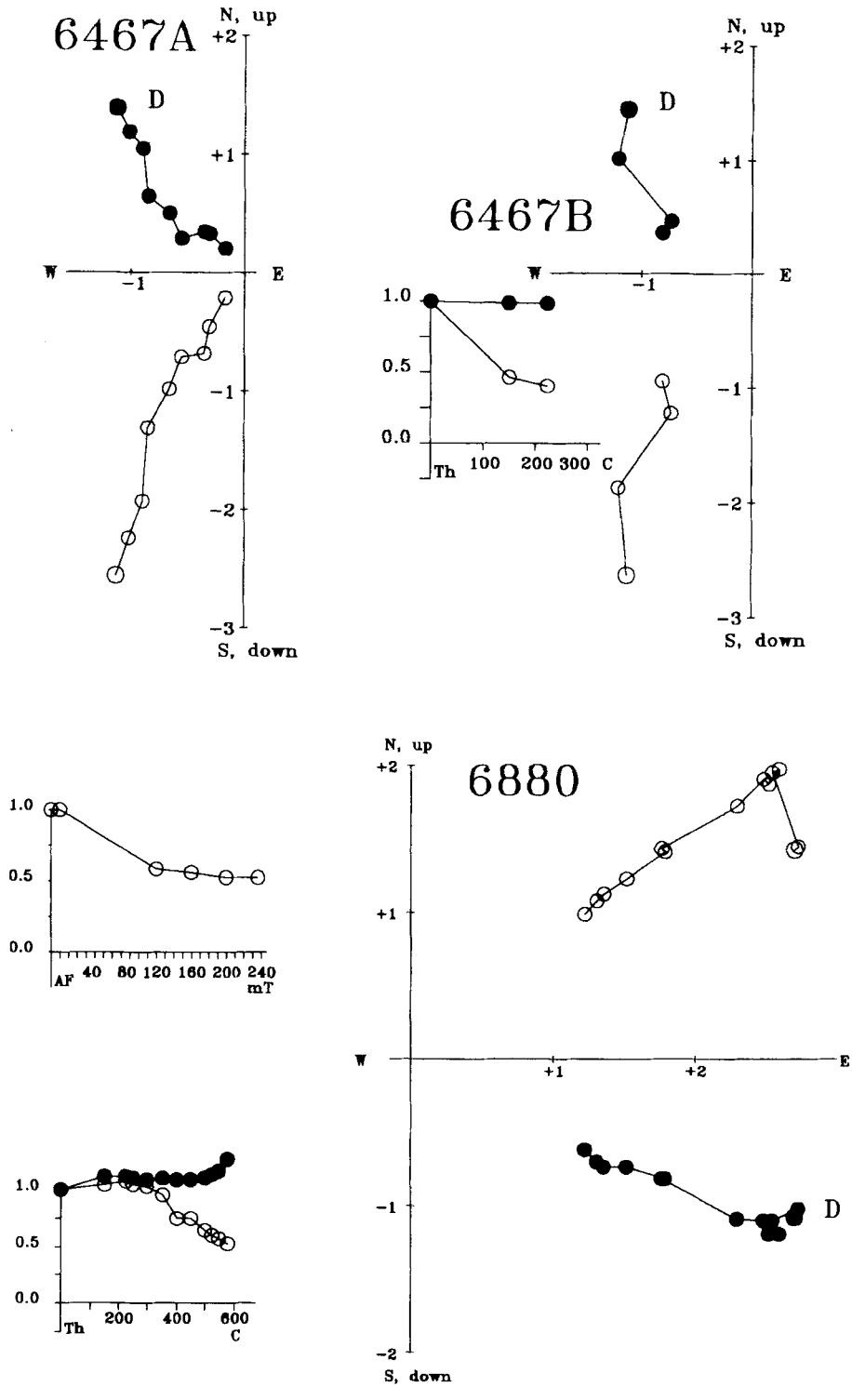


Figure 4. *(Continued.)*

either diamagnetic or paramagnetic/ferromagnetic, but had a very low value (maximum 2×10^{-5} SI). Initial remanence intensities were typically about $10^{-5}-10^{-4}$ A m^{-1} in 'C' members. In 'A' and 'B' members the intensity was higher, typically about 10^{-4} A m^{-1}, and occasionally as high as a few times 10^{-3} A m^{-1}.

The correlation between susceptibility and remanence intensity is poor, due to the complex magnetic mineralogy and the existence of both normal- and reversed-polarity remanence, with variable amounts of overprint.

In the Agostyán Gorba quarry section (Fig. 2) the 'C' members are often disturbed by fissures and microkarstic cavities. Thus, due to the higher clay content, the magnetic properties of several samples from 'C' members were not very different from those drilled in 'A' or 'B' members at this locality.

In all members, goethite, magnetite and haematite occur

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together (Fig. 3), although their proportion in different samples is very different. On heating, the soft component, which is sometimes dominant (e.g. specimen 6467A) or becomes dominant after the removal of the remanence of the goethite at a maximum of 150 °C (e.g. specimen 6465A), invariably decays gradually and disappears at the Curie point of magnetite. There are samples with dominant medium and hard components, which persist above the Curie point of magnetite (e.g. specimens 6476A and 6905B), i.e. the remanence most probably resides in haematite.

Concerning the actual carriers of the natural remanence, the behaviour of the NRM and the susceptibility on demagnetization suggest magnetite with or without a contribution from goethite in pure 'C' members, with the addition of variable amounts of haematite in 'A' and 'B' members (Fig. 4).

The block diagrams of the remanence intensities lost between demagnetization steps (Fig. 5) reveal common features, as well as important differences, between the 'A', 'B' and 'C' members. The most important common feature is that goethite may occur as the dominant carrier of the remanence, as is evident from the behaviour of the NRM on combined AF and thermal demagnetization: compared to the natural state, the loss in intensity at AF 10 mT is negligible, while that between AF 10 mT and 150 °C is often dramatic. The most striking difference between members is that in 'C' members the unblocking of the NRM is complete by 400 °C, while in the other members the NRM persists to higher temperatures.

Goethite may carry a larger or smaller part of the NRM, but its share in the NRM has no relation to polarity. Polarity is not dependent on the contribution of haematite to the NRM either (compare e.g. specimens 6635A and 6879 in Fig. 4). Moreover, the direction carried by haematite coincides with that of the component with the lower blocking temperature. The remanence direction of the goethite may also be aligned with those of the magnetite and haematite. More often, however, goethite is of overprint character, i.e. part of it at least is secondary (e.g. the mean direction of the well-defined subtracted vectors between NRM or AF 10 mT and TH 150 °C of Fig. 5 is $D=357°$, $I=58°$, $k=13$, $\alpha_{95}=16°$, $n=8$).

After the removal of the overprint (usually from the third demagnetization step), a characteristic magnetization is obtained for a vast majority of the samples. Exceptions are three samples from Lábatlan, in zone A_7 (Fig. 7) which become very weak at 150°, and four samples in the rip-up breccia

Table 1. Comparison of palaeomagnetic locality mean directions for Lábatlan and Agostyán. N: number of samples; D°, I° and $D_c°$, $I_c°$: mean palaeomagnetic declination, inclination, before and after tilt correction; k and α_{95}: statistical parameters (Fisher 1953). Angular separation of N + R vectors is 10° for Lábatlan and 10.8° for Agostyán. The combination of data for normal and reversed polarity groups and their mean is described in the text. The mean of all samples was calculated from all independently oriented samples with single-component remanence.

	N	D°	I°	k	α_{95}	$D_c°$	$I_c°$
Lábatlan lat: 47.8 lon: 18.5							
normal polarity	35	306	48	62	3		
reversed polarity	10	112	-47	65	6		
mean of normal and reversed polarity groups	2	299	48			306	54
all samples	58	306	47	43	3	313	52
Agostyán lat: 47.6 lon: 18.5							
normal polarity	16	310	53	62	4		
reversed polarity	6	107	-51	78	6		
mean of normal and reversed polarity groups	2	298	53			305	53
all samples	30	300	52	26	5	307	52

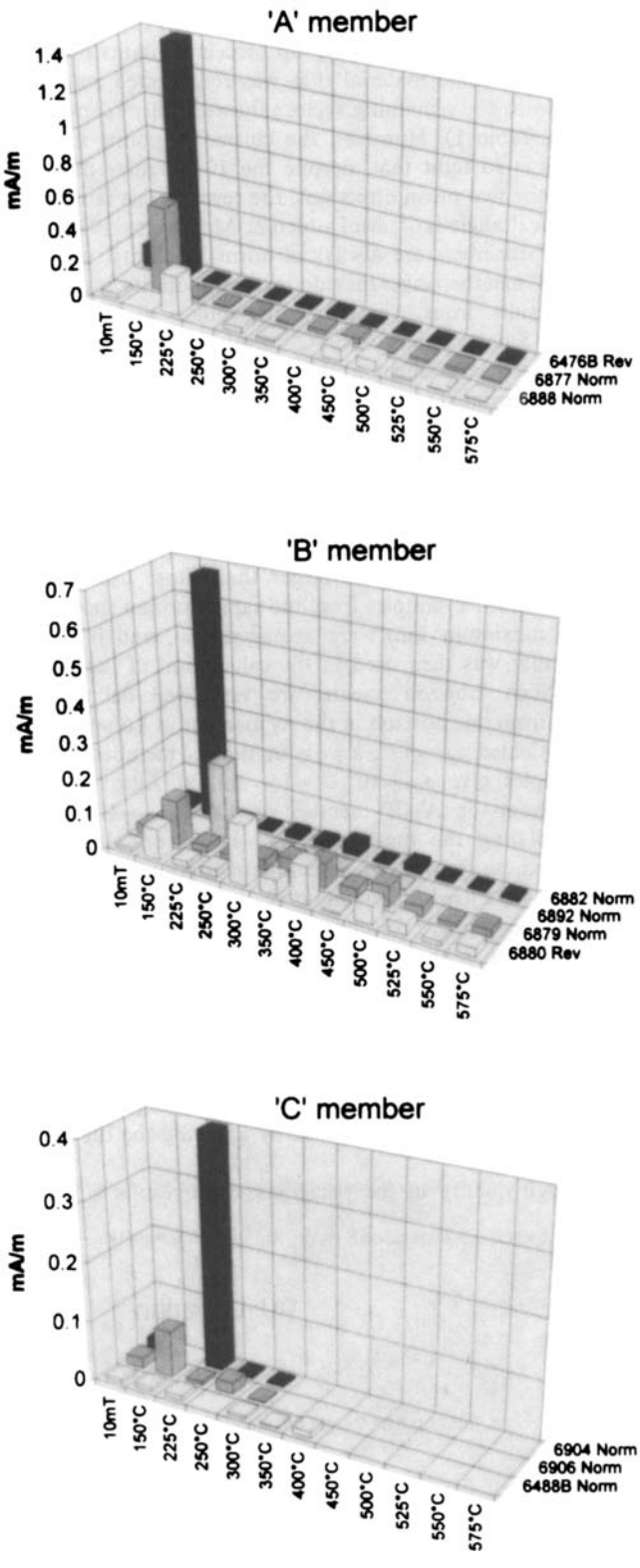


Figure 5. Lábatlan. Intensity of subtracted vectors of the NRM. Demagnetization steps: 0–12 NRM, AF 10 mT, TH 150, 225, 250, 300, 350, 400, 450, 500, 525, 575.

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horizon. This latter may be regarded as a kind of positive conglomerate test, since the rip-up breccia contains reworked platform-carbonate material that was redeposited soon after the breakup. The remaining vectors show excellent consistency (Fig. 6a, Table 1). However, the cluster of points for both polarities is so tight that, despite the 10° angular difference between the two mean directions, the reversal test is negative. (The critical angle is 7°; McFadden & McElhinny 1990.)

The 'C' members are always of normal polarity, while the 'A' and 'B' members also include reversed polarity samples.

The results from Agostyán are similar to those from Lábatlan. Some samples from 'C' members become too weak for reliable measurement at 150 °C. Six samples are clearly slightly overprinted (or show transitional directions?). The rest are shown in Fig. 6(b) and in Table 1. Once again, the reversal test is negative (the angular difference between the mean directions of the normal- and reversed-polarity samples is 10.8°; the critical angle is 7°).

It is noted that the numbers of good samples at both Lábatlan and Agostyán are larger than those used for the reversal test, since samples from the same horizon (horizontal distance maximum 1 m) were averaged first, and it is their average that was then used in the calculation of the locality means. Such 'horizon' means are very well defined. For instance, from one horizon in the A_2 member at Lábatlan, four samples yielded $\alpha_{95}=12°$, $k=61$; in the A_4 member another four samples give $\alpha_{95}=10°$, $k=86$; and at Agostyán, four samples from the 'A'–'B' member provided $\alpha_{95}=10°$, $k=80$. For calculating the mean direction based on all samples (Table 1), those showing single-component NRM after demagnetization were used individually.

The stereographic projections (Fig. 6), as well as the VGP latitudes (Figs 7 and 8), illustrate the high quality of the results, especially for Lábatlan. On tilt correction, the mean directions for Lábatlan and Agostyán move closer (Table 1). Considering the very low tilt angle (at Lábatlan, it is 8°; at Agostyán, it is 5°), it is not practical to apply a rigorous fold test. Nevertheless, it is worth noting that the angular distance between locality means is reduced by applying the tilt corrections (Table 1).

The high quality of the palaeomagnetic results is further supported by the consistent VGP latitude pattern in two parallel sections in the Lábatlan quarry, about 60 m apart (Figs 7b and c).

An exact correlation between the Lábatlan and Agostyán sections is not possible on stratigraphic grounds. However, both sections are of Rhaetian age, and are considered to be older than latest Rhaetian.

It is possible, therefore, that the reversals observed at Lábatlan may correlate with some of those defined in the Agostyán section. This may be relevant to magnetostratigraphy, which is beyond the scope of this study. Nevertheless, it should be mentioned that the recently published Mesozoic polarity time-scale of Gradstein *et al.* (1994) shows reversed polarity with two short normal events for the part of the Rhaetian in which the Lábatlan and Agostyán sections are situated. It should also be noted that in the Lábatlan quarry, above the condensed (probably 4th order) interval sampled (Fig. 7b), there are 11 other cycles, visible, although less condensed. Thus we predict that the polarity time-scale for the Rhaetian proposed by Gradstein *et al.* (1994) will have to be modified.

6 CONCLUSIONS

Our methodological study on platform carbonates with well-developed Lofer cyclicity has demonstrated that such rocks are likely to preserve nearly continuous records of the ancient magnetic field through geological periods. The reason for this may be that general circulation of ground water may be prevented by frequent intercalation of the argillaceous 'A' horizons of palaeosol origin.

Lofer cyclicity offers an excellent means to calibrate the magnetic-polarity time-scale, if we take into account the condensed intervals and the possibility of the 'missing beats' that may be assumed in composite palaeosol layers. Polarity changes within 'A' and 'B' members or at their boundaries suggest long time gaps (sequence boundaries) at certain places in the carbonate platform successions.

In the Lábatlan section, polarity changes were documented within 'A' and 'B' members, but never in 'C' members. In

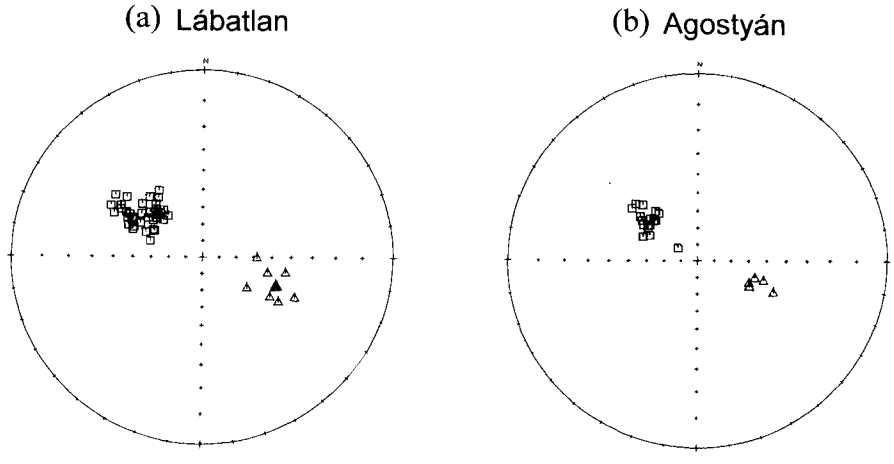


Figure 6. Palaeomagnetic declination–inclination plot for (a) Lábatlan and (b) Agostyán. Stereographic projection. Squares are positive inclinations; triangles are negative inclinations.

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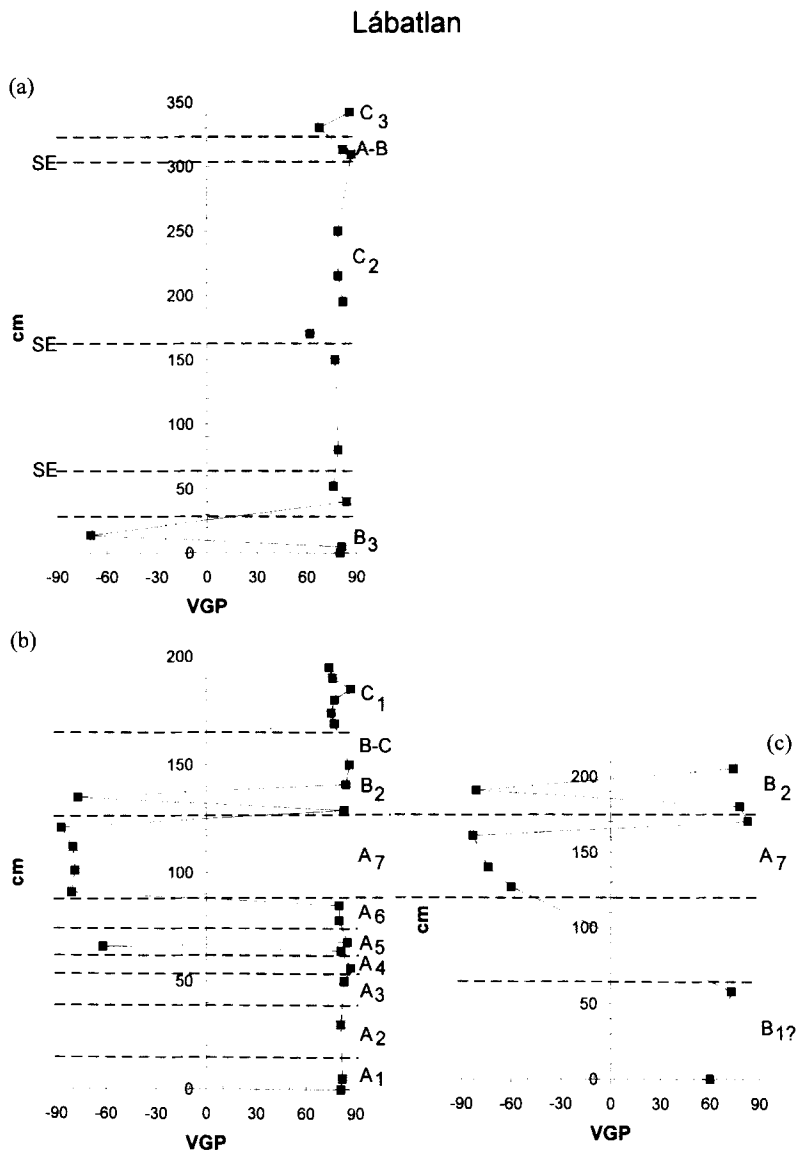


Figure 7. Lábatlan. VGP latitudes and members of the Lofer cycles. All surfaces between members (horizontal dashed lines) are disconformity surfaces, except those marked with SE (submarine erosion surface).

$A_{(1-7)}$: argillaceous limestone layers of intertidal–subtidal facies, altered by subaerial weathering and soilification. Rootcasts are common in layers A_4, A_6 and A_7.

B_1, B_2, B_3: intertidal algal-mat facies.

B–C: rip-up breccias.

C_1–C_3: subtidal, skeletal packstone and wackestone with *Megalodonts* in C_1 and C_3.

A–B: transitional facies between A and B.

B–C: transitional facies between B and C.

The horizontal distance between sections (b) and (c) is about 60 m. Section (a) starts about 10 m above the top of section (b).

contrast, reversals occur at Agostyán within each member. Fissures and karstic cavities are frequent at Agostyán, i.e. part of the original 'C' member was replaced by material deposited later. Therefore, we conclude that sections even with this type of 'primary' alteration are best avoided in magnetostratigraphy work.

Our first results on Late Triassic platform carbonates indicate that a sampling strategy based on knowledge of the sedimentology, tectonics, etc. combined with suitable laboratory analysis of the magnetic properties may lead to a reliable polarity time-scale for a period of time not yet calibrated adequately for polarity reversals. Furthermore, tectonic applications are not handicapped by syn- or early post-diagenetic karstification. Triassic platform carbonates are widespread in the Southern Alps, Eastern Alps and the Carpathian–Pannonian region. Therefore, palaeomagnetic directions from geographically distributed sections should be of considerable tectonic importance.

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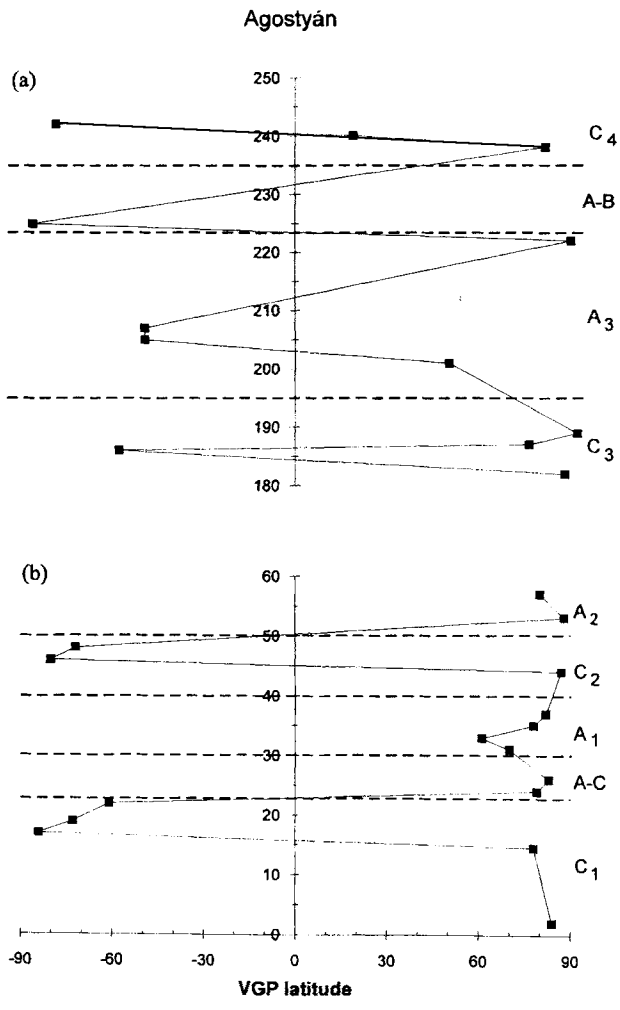


Figure 8. Agostyán. VGP latitudes and members of the Lofer cycles. All surfaces between members are disconformity surfaces. Fissures and karstic cavities are abundant in the C_1 member, below the A–C layer, and the C_3 member, below A_3.

A_{1-3}: argillaceous limestone of greenish colour; reworked palaeosol.

A–B: microlaminated limestone, intermediate between A and B.

$C_{1,2,4}$: subtidal skeletal packstone and wackestone.

The horizontal distance between sections (a) and (b) is 6 m.

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